

Asymptotic statistics on stratified spaces

Lecture 4: Limiting laws on stratified spaces; open questions

Slides available at github.com/statdivlab/asymptotics-stratified-spaces

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Recap

Definition: A *stratified space* is a metric space S with a finite partition into nonempty *strata* $S = \bigsqcup_j S_j$ such that:

1. each S_j is a boundaryless manifold of dimension s_j .
2. For $j \neq k$: $\overline{S_j} \cap S_k \neq \emptyset \implies S_k \subseteq \overline{S_j}$ and $s_k < s_j$.

► Three stratified metric spaces

- K -spider S_K
- Open book $\mathcal{O}$ for fixed d and K
- $\mathcal{T}_{m+2}$

Recap

Definition: A complete metric space (S, d) is *globally nonpositively curved* (NPC, CAT(0)) if for all $x, y \in S$ there is $m \in S$ such that for every $z \in S$,

$$d^2(z, m) \leq \frac{1}{2}d^2(z, x) + \frac{1}{2}d^2(z, y) - \frac{1}{4}d^2(x, y).$$

- ▶ CAT(0) spaces have unique geodesics
- ▶ CAT(0) spaces have unique Fréchet means
- ▶ Three CAT(0) spaces
 - ▶ K -spider S_K
 - ▶ Open book $\mathcal{O}$ for fixed d and K
 - ▶ $\mathcal{T}_{m+2}$

Recap

- ▶ Not all stratified spaces are CAT(0)
 - ▶ eg a positively curved manifold
- ▶ Phenomenon in stratified and CAT(0) spaces: *sticky* Fréchet means
 - ▶ $\mu(P)$ doesn't change stratum when P is perturbed

What are limiting distributions for Fréchet means on stratified spaces?

Limiting distributions

On stratified NPC spaces, we have unique and consistent Fréchet means.

To talk about distributions, we need a *chart*:

$G \subseteq S$, $\mu \in G$, a homeomorphism $\phi : G \rightarrow U \subseteq \mathbb{R}^s$ for U open.

Then, we'll study $\sqrt{n}(\phi(\hat{\mu}_n) - \phi(\mu))$.

Limiting distributions

If μ is in stratum S_j , consider choosing

- ▶ $G \subseteq S_j$ a neighbourhood of μ
- ▶ $\phi = \log_\mu$

and use

$$\sqrt{n}(\phi(\hat{\mu}_n) - \phi(\mu)) \xrightarrow{\mathcal{D}} N_s(0, \Lambda^{-1}\Sigma\Lambda^{-1}),$$

where, for $m_\theta(x) = d^2(\phi^{-1}(\theta), x)$,

- ▶ $\Sigma = \text{Var}(\dot{m}_{\theta_0}(X_1))$
- ▶ $\Lambda = P\ddot{m}_{\theta_0}$

Let's talk about $\phi = \log_\mu$ for stratified spaces.

CLTs with μ on a top-dimensional stratum: Open books

The “easy” case: μ is in a top-dimensional stratum

In the open book, non-sticky case $\mu \in L_k^+$, the log map is the folding map $\Phi_k : \mathcal{O} \rightarrow \mathbb{R}^{d+1}$.

Let P be nondegenerate with $F(\mathbf{0}) < \infty$. If $m_k > 0$:

$$\sqrt{n}(\Phi_k \hat{\mu}_n - \Phi_k \mu) \xrightarrow{\mathcal{D}} N_{d+1}(0, \text{Var}(\Phi_k X))$$

Result due to Hotz et al (2013)

CLTs with μ on a top-dimensional stratum: Tree space

What is $\log_\mu$ on an open book when $\mu \in \mathcal{T}_{m+2}$ is binary?

The geodesic from μ to x is a straight segment in the orthant of μ , then a sequence of straight segments through neighbouring orthants.

Write $\log_\mu x = d(\mu, x) v(x) \in \mathbb{R}^m$ with $v(x)$ the initial direction, and for $t^* \in \mathbb{R}_{>0}^m$ the coordinates of μ ,

$$\ell_\mu(x) := t^* + \log_\mu x \in \mathbb{R}^m$$

CLTs with μ on a top-dimensional stratum: Tree space

$$\ell_\mu(x) := t^* + \log_\mu x \in \mathbb{R}^m$$

The algebraic form of $\mu \mapsto \log_\mu x$ is locally constant in μ for a top-dimensional μ

Let $D_\mu \subseteq \mathcal{T}_{m+2}$ be the set of x whose geodesic from μ passes through a stratum of codimension ≥ 2

Lemma: (Barden, Le & Owen 2018) For $x \notin D_\mu$, $\nu \mapsto \ell_\nu(x)$ is differentiable at μ with derivative $M_\mu(x) \in \mathbb{R}^{m \times m}$, nsd. Moreover $M_\mu(x) = 0$ unless $x \in \mathcal{S}_\mu$, the set of x whose geodesic from μ bends.

CLTs with μ on a top-dimensional stratum: Tree space

Theorem: (Barden, Le & Owen 2018) Let P on $\mathcal{T}_{m+2}$ have $F < \infty$ and Fréchet mean μ in a top-dimensional stratum, with $P(D_\mu) = 0$. Let $\hat{t}_n$ be the coordinates of $\hat{\mu}_n$. Then

$$\sqrt{n}(\hat{t}_n - t^*) \xrightarrow{\mathcal{D}} N_m(0, A^{-1}VA^{-1})$$

where

$$V = \text{Var}(\log_\mu X_1)$$

and

$$A = I - \mathbb{E} M_\mu(X_1)$$

for $M_\mu(x) \in \mathbb{R}^{m \times m}$ the derivative in ν of the translated log map, $\nu \mapsto \ell_\nu(x)$

CLTs with μ on a top-dimensional stratum: Tree space

Previously: $\Sigma = \text{Var}(\dot{m}_{\theta_0}(X_1))$ and $\Lambda = P\ddot{m}_{\theta_0}$ and

$$\sqrt{n}(\phi(\hat{\mu}_n) - \phi(\mu)) \xrightarrow{\mathcal{D}} N_s(0, \Lambda^{-1}\Sigma\Lambda^{-1})$$

Here, $V = \text{Var}(\log_{\mu} X_1)$ and $A = I - \mathbb{E} M_{\mu}(X_1)$. and

$$\sqrt{n}(\hat{t}_n - t^*) \xrightarrow{\mathcal{D}} N_m(0, A^{-1}VA^{-1})$$

- ▶ Locally, $\nabla_t d^2(\phi^{-1}(t), x) \Big|_{t=t_{\nu}} = -2\log_{\nu}(x)$
- ▶ $\dot{m}_{\theta}(x) = 2\theta - 2\ell_{\phi^{-1}(\theta)}(x)$
 - ▶ Use fact μ in a top-dimensional stratum iff $\int \log_{\mu} x dP(x) = 0$.
- ▶ For $x \notin D_{\phi^{-1}(\theta)}$, $\ddot{m}_{\theta}(x) = 2I - 2M_{\phi^{-1}(\theta)}(x)$

$P(D_{\mu}) = 0$ plays the role of the **support condition**.

CLTs with μ on a co-dimensional stratum: Tree space

Barden, Le & Owen 2018 gave the limiting behaviour when μ lies in a codimension-1 stratum, for arbitrary dimension.

Barden & Le followed up with a study of log maps on *orthant spaces* ($\subset$ stratified spaces), thereby giving the needed pieces for general CLTs on orthant spaces, including and tree space.

Let's look at the open book case.

CLTs with μ on a co-dimensional stratum: Open books

$S = \mathcal{O}$. What about when μ is in a co-dimensional stratum?

- ▶ Sticky case, $\mu \in L_0$: the map is the projection onto the spine
 $P_S(x^{(0)}, \tilde{x}, k) = \tilde{x}$.
- ▶ Partly sticky, $\mu \in L_0$, $m_k = 0$: the map is the folding map onto orthant k followed by the convex projection onto $L_0 \cap L_k$:
 $\hat{P} \circ \Phi_k$ for

$$\hat{P}(x^{(0)}, \tilde{x}) = (\max\{x^{(0)}, 0\}, \tilde{x}),$$

(negative part collapsed onto the spine)

The chart changes with the measure? The support of the distribution changes with the measure?

CLTs with μ on a co-dimensional stratum

What's the chart?

The chart is $\log_\mu : S \mapsto T_\mu S$.

However, this isn't enough to understand the limit of $\sqrt{n}\log_\mu \hat{\mu}_n$, because tangent spaces of stratified spaces may not be linear spaces.

CLTs with μ on a co-dimensional stratum

A general framework for CLTs on *stratified spaces* was missing until Mattingly, Miller and Tran released a sequence of preprints in 2023.

The CLT for stratified spaces

Things to know about the CLT for stratified spaces:

What plays the role of ϕ ?

- ▶ $\mathcal{L} \circ \log_\mu$
- ▶ $S \xrightarrow{\log_\mu} T_\mu S \xrightarrow{\mathcal{L}} \mathbb{R}^m$
- ▶ $\mathcal{L}$ is a *tangential collapse*
 - ▶ folds $T_\mu S$ into a vector space (think: open book case)
 - ▶ a continuous, angle-preserving function with the “right” mean ($L_\#(\log_\mu)_\# P = 0$)

The CLT for stratified spaces

What's the rate?

- ▶ $\sqrt{n}$, because of restrictions on P that exclude singular Hessians

The CLT for stratified spaces

Where does the limiting distribution have support?

- ▶ Not on the tangent space $T_\mu S$, which may not be linear
- ▶ Not on the *escape cone* (set of points in $T_\mu S$ where $\nabla_\mu F(X) = 0$), because the support isn't right
- ▶ The *fluctuating cone*

The CLT for stratified spaces

What does the limiting distribution look like?

- ▶ a normal, pushed forward in two ways

How can we understand its variance?

- ▶ like the usual $\text{Var}(\dot{m}_{\theta_0})$, but with the effect of the pushforward wrapped in

CLTs for stratified spaces

Theorem: Under conditions,

$$\sqrt{n} \log_{\mu} \hat{\mu}_n \xrightarrow{\mathcal{D}} \mathcal{H}_{\sharp} N(0, \Sigma),$$

where $\Sigma = \text{Var}((\mathcal{L} \circ \log_{\mu})_{\sharp} X_1)$ and

- ▶ $\log_{\mu} : S \rightarrow T_{\mu}S$. $T_{\mu}S$ is a cone; $T_{\mu}S$ is not nec a vector space
- ▶ $\mathcal{L}$ is *tangential collapse*. It maps the tangent space $T_{\mu}S$ into $\mathbb{R}^m$
 - ▶ $\mathcal{L} = \text{id}$ on a manifold, $\mathcal{L} = \Phi_k$ on $\mathcal{O}$
- ▶ $\mathcal{E}$ is the *escape vector*: the influence function of μ under P
 - ▶ On $\mathbb{R}^m$: $\mathcal{E}(\delta_x) = x - \mu$; on a manifold: $(\text{Hess}_{\mu} F)^{-1}(\log_{\mu} x)$
- ▶ $\mathcal{H} = \mathcal{E} \circ \mathcal{L}^{-1}$ is the *distortion map*
 - ▶ $\mathcal{H} : \mathbb{R}^{\ell} \rightarrow \overline{\mathcal{C}_P} \subseteq T_{\mu}S$
 - ▶ $\mathcal{H}(V)$ for $V \in \mathbb{R}^{\ell}$, pick any finitely supported Δ on $\text{supp}(\log_{\mu})_{\sharp} P$ with $\int \mathcal{L} d\Delta = V$ and take its escape vector

CLTs for stratified spaces

It applies

- ▶ For *smoothly stratified metric spaces*
 - ▶ like our definition from yesterday, but also. . .
 - ▶ complete, geodesic, locally compact $\text{CAT}(\kappa)$ space
 - ▶ every point has a local exponential map: $\forall p \in M \exists \varepsilon > 0$ s.t. $\log_p$ restricted to $B(p, \varepsilon)$ is a homeomorphism onto its image
- ▶ For localized^{***}, immured^{**}, amenable^{*} distributions
 - ▶ ^{*} gives “nice” Taylor expansion of Frechet function
 - ▶ excludes degenerate Hessians (thus, smeariness)
 - ▶ plays the role of $P(D_\mu) = 0$ (?)
 - ▶ ^{**} gives $\mathcal{E}(\Delta) \in \overline{C_P}$.
 - ▶ ^{***} needed for tangential collapse to exist

The escape cone

Definition: The *escape cone*, *hull* and *fluctuating cone* of P are

$$E_P = \{X \in T_\mu S : \nabla_\mu F(X) = 0\},$$

$$\text{hull } P = \text{hull supp}(\log_\mu)_\# P$$

$$C_P = E_P \cap \text{hull } P.$$

Escape cone = where escape vectors can live

Stickiness is about the escape cone

$S = \mathcal{O}$: On the closed leaf L_k the space is flat and $F|_{L_k} = \ell_k \circ \Phi_k$ for a Euclidean Fréchet function ℓ_k . Hence $\nabla_\mu F$ is *linear* on L_k . For V_k is the unit tangent vector at $\mu \in L_0$ that points straight off the spine into leaf k :

$$V_k = (1, \mathbf{0}_d, k) \in T_\mu \mathcal{O},$$

we have that

$$\nabla_\mu F(V_k) = -m_k.$$

So

sticky	$m_j < 0 \ \forall j$	$E_P = L_0$
partly sticky	$m_k = 0$	$E_P = L_k$
nonsticky	$m_k > 0$	$E_P = T_\mu \mathcal{O} = \mathbb{R}^{d+1}$

Connecting to easier cases

In $\mathbb{R}^m$ and on a manifold, $\nabla_\mu F = 0$ defines the tangent space, so $E_P = T_\mu S$.

Similarly, in the stratified setting, if μ lies in a maximal stratum (BP setting!) then

- ▶ $E_P = T_\mu S$ is a vector space
- ▶ $C_P = \mathbb{R}^\ell$
- ▶ $L = \text{id}$
- ▶ $\mathcal{H} = \mathcal{E} \circ L^{-1} = \mathcal{E}$, and $\mathcal{E}(\Delta) = (\text{Hess}_\mu F)^{-1} \Delta$,

Then $\mathcal{H}_\# N(0, \Sigma) = N(0, (\text{Hess}_\mu F)^{-1} \Sigma (\text{Hess}_\mu F)^{-1})$

Summary

- ▶ Mattingly, Miller and Tran did beautiful work to construct a CLT for stratified spaces with curvature bounded above

$$\sqrt{n} \log_{\mu} \hat{\mu}_n \xrightarrow{\mathcal{D}} \mathcal{H}_{\sharp} N(0, \Sigma),$$

where $\Sigma = \text{Var}((\mathcal{L} \circ \log_{\mu})_{\sharp} X_1)$

- ▶ You can think of it as a fusion of the “omnibus”/chart CLT (BP) and the orthant spaces CLT (Barden & Le)
- ▶ $\mathcal{L}$ is *tangential collapse* deals with non-vector space tangent space
 - ▶ $\mathcal{E}$ is the *escape vector*, concerning the influence of points
 - ▶ $\mathcal{H} = \mathcal{E} \circ \mathcal{L}^{-1}$ is the *distortion map*

Active work: Towering space

My expertise is in statistical methods for the analysis of microbiome/biodiversity data. I think a lot about phylogenetic trees and what other people call “compositional” data.

I have never, ever seen an example of bacterial trees that were not missing leaves.

e.g., $T_1 \in \mathcal{T}_{\{A,B,C,D\}}$, $T_2 \in \mathcal{T}_{\{A,B,C,D,E\}}$, $T_3 \in \mathcal{T}_{\{A,B,C,D,F\}}$,
 $T_4 \in \mathcal{T}_{\{A,B,C,D,E,G\}}$

i.e., I've never seen truly BHV-valued data in my applied work.

Active work: Towering space

Let $\mathcal{N}$ be the union of all (possible) leaves

An equivalence relation between $T_1 \in \mathcal{T}^{\mathcal{L}_1}$ and $T_2 \in \mathcal{T}^{\mathcal{L}_2}$: $T_1 \simeq T_2$, if there is a finite number of prunings and regraftings that can be applied sequentially to T_1 that result in T_2 .

- ▶ pruning: remove a leaf from an internal node
- ▶ regrafting: add a leaf at an internal node

Active work: Towering space

Why exclude regrafting leaves from anywhere? Paradoxes.

Active work: Towering space

The **Towering space** is the metric space $(\mathcal{T}^{\mathcal{P}(\mathcal{N})} / \simeq, d)$, where

- ▶ $\mathcal{T}^{\mathcal{P}(\mathcal{N})} = \bigcup_{\mathcal{L} \subseteq \mathcal{N}, |\mathcal{L}| \geq 3} \mathcal{T}^{\mathcal{L}}$ is the union of all BHV tree spaces over all subsets of possible leaves
- ▶ d is the quotient pseudometric induced by the BHV distance

Active work: Towering space

This is *very open* and *very important*. I have a huge number of datasets in hand with $T_1 \dots T_n \in \mathcal{T}^{\mathcal{P}(\mathcal{N})}$

What do we know about towering space? *Some properties of geodesics*

What do we know about the geometry and topology of towering space? *Almost nothing*

Closing comments

Here's my mental model of what to work on:

- ▶ What question does the scientist have?
- ▶ What is the randomness or uncertainty mechanism?
- ▶ What data does it generate? What's its sample space?
- ▶ What's a meaningful parameter here? What's its parameter space?
- ▶ Is the parameter identifiable? Under what assumptions?
- ▶ Do we have estimators for it? What do we know about these estimators? Under what assumptions?
- ▶ Are new mathematical or statistical tools needed to answer #1?

Closing comments

I feel that geometric statistics has been stuck at “What’s a meaningful parameter?”

Fréchet means are only one parameter; squared loss minimizers have been a huge focus.

Overlap? Let's talk!

My attendance this program is motivated in part by the following questions/topics:

- ▶ Towering space and its properties
- ▶ Any tree that I am using is estimated – from sequence data.
It is a solution to some optimization problem.
- ▶ Escape vectors as influence functions. Variance bounds?
- ▶ Model / objective misspecification for non-Euclidean parameters
 - ▶ improved estimators of parameters (a la OSE/TMLE, connects to IF interest)
 - ▶ improved estimators of variances (a la sandwich estimation)
- ▶ ~~"wrong" metric space~~ Identifiability issues
 - ▶ long branch attraction

Overlap? Let's talk!

If you are interested in any of these questions, and/or care about developing methods for the analysis of sequence data (esp. with phylogenetic trees), let's talk more!

References

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Smoothly stratified metric spaces

Definition: (MMT 2023, Def. 2.12) A *smoothly stratified metric space* is a complete, geodesic, locally compact $\text{CAT}(\kappa)$ space with a decomposition

$$M = \bigsqcup_{j=0}^d M^j \quad \text{into locally closed strata with} \quad \overline{M^j} = \bigcup_{k \leq j} M^k,$$

such that

1. (*manifold strata*) each $(M^j, d|_{M^j})$ is a smooth manifold whose geodesic distance is the restriction $d|_{M^j}$;
2. (*local exponential maps*) for each $p \in M$ there is $\varepsilon > 0$ such that $\log_p$ restricted to $B(p, \varepsilon)$ is a homeomorphism onto its image.

Escape vectors

Theorem: (Prop. 4.34, Thm. 4.37) Let P be amenable, $\Delta_n \Rightarrow \Delta$ sampled from $T_\mu M$, $t_n \downarrow 0$. Then $\mathcal{E}(\Delta)$ is well defined,

$$\mathcal{E}(\Delta) = \lim_{t \rightarrow 0} \frac{1}{t} \arg \min_{X \in E_P} \left(F(\exp_\mu X) - t \langle \Delta, X \rangle \right)$$

Definition: For $\delta \in M$, the *escape vector* is

$$\mathcal{E}(\delta) = \lim_{t \rightarrow 0^+} \frac{1}{t} \log_\mu \overline{P + t\delta},$$

extended to $\Delta \in T_\mu M$ by $\mathcal{E}(\Delta) = \mathcal{E}(\exp_\mu \Delta)$. For $\Delta = \sum_i \lambda_i \delta_{Y^i}$ set

$\delta \in M$ is a *finitely supported* measure

Existence is a theorem!

misc reminders to self

- ▶ $L = \text{id}$ on a manifold (and $\mathbb{R}^\ell = \mathbb{R}^m$ if P doesn't collapse along subspaces)

A blank slide!

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